# Radar Telemetry Over LoRa

Project Progress Update
Manuel Medina
WES-207

# Project Goal

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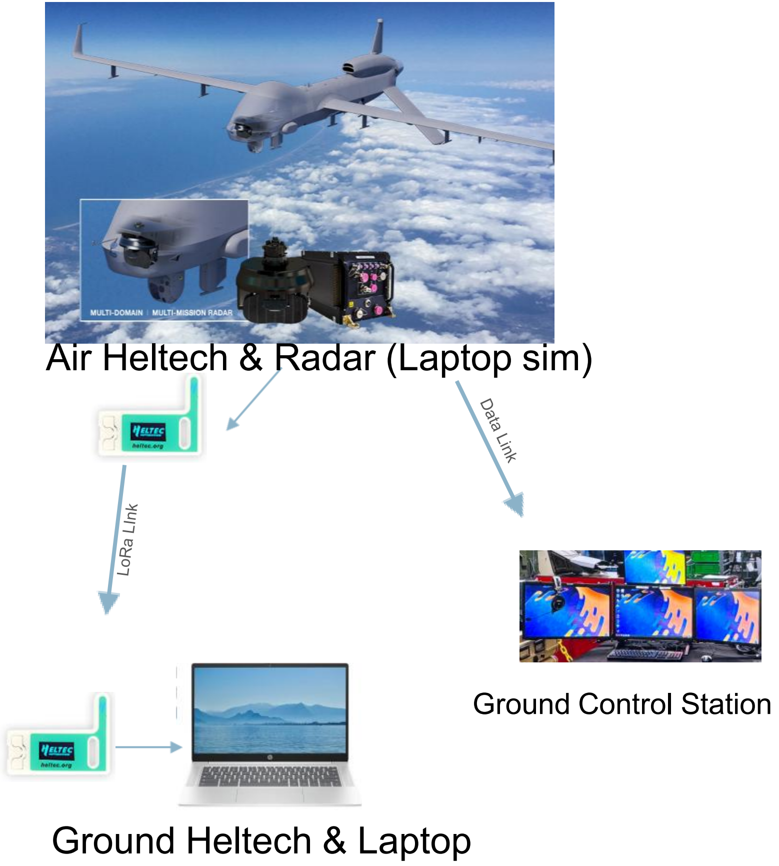
- Build a wireless radar test link between an air side and a ground side
- Let the ground side send commands
- Let the air side return updated radar status
- Will help determine whether a problem is in the radar or in the ground control station network

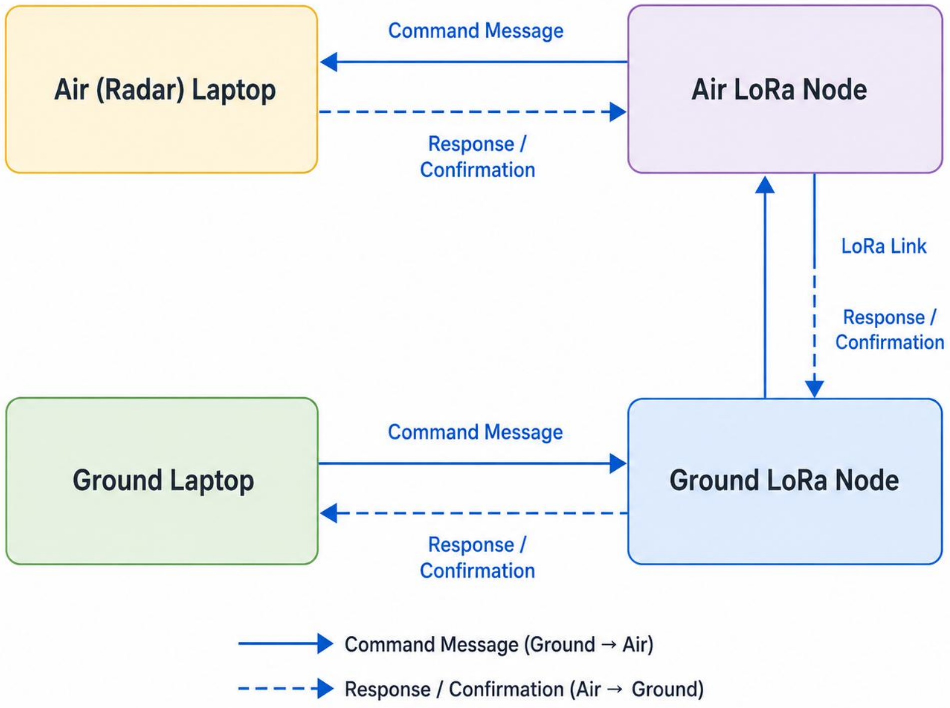
**Simple Explanation**

- The ground laptop sends a command
- The air laptop receives it
- The air laptop sends the updated radar status back
- Both sides show the data on screen



# System Setup

- 2 Laptops
- 2 Heltec LoRa boards
- 2 Heltec LoRa boards
- Ground Heltec sends commands over LoRa
- Air Heltec forwards commands to the air laptop
- Air laptop sends updated radar status back
- Ground laptop shows and logs the returned status



# Modes Being Tested & Test Card

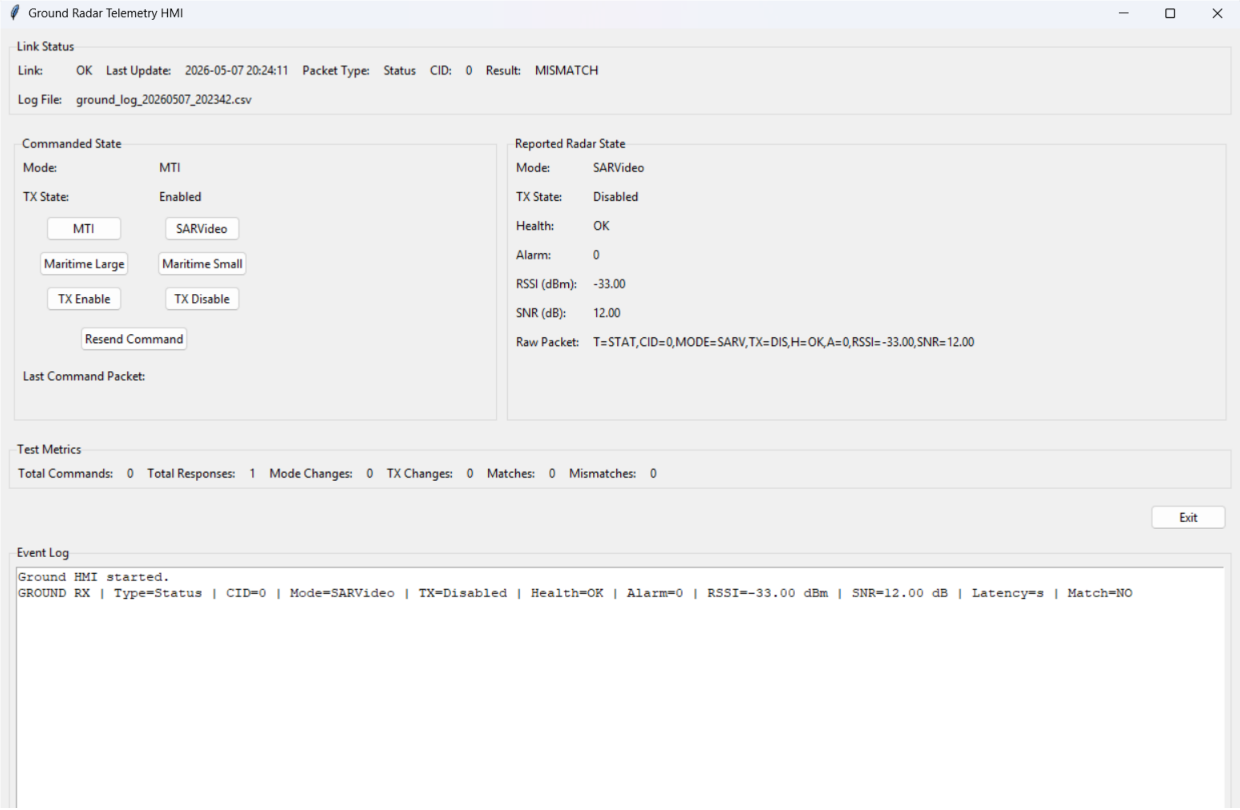
**Radar Modes**

- Motion Target Indicator (MTI)
- SARVideo
- Maritime Large Vessel
- Maritime Small Vessel
- TX Enable
- TX Disable

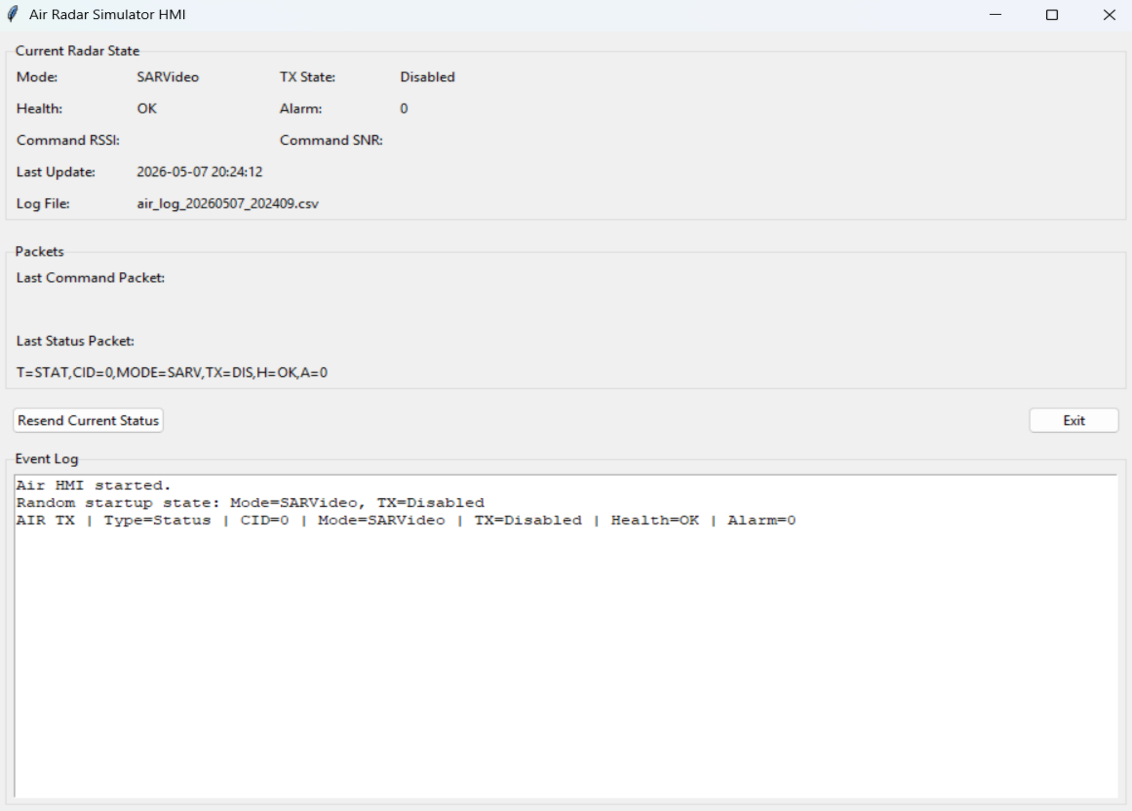
**Test Card**

- Send commands from the ground laptop
- Check that the air laptop updates correctly
- Check that the ground laptop receives the correct reply
- Record success or failure for each test
- Static test from 0ft to 200ft at increments of 20 feet.
- Slow speed (10 to 20 MPH) driving test

# Ground HMI


Ground Radar Telemetry HMI
Link Status
Link: OK Last Update: 2026-05-07 20:24:11 Packet Type: Status CID: 0 Result: MISMATCH
Log File: ground_log_20260507_202342.csv
Commanded State
Mode: MTI
TX State: Enabled
MTI
SARVideo
Maritime Large
Maritime Small
TX Enable
TX Disable
Resend Command
Last Command Packet:
Reported Radar State
Mode: SARVideo
TX State: Disabled
Health: OK
Alarm: 0
RSSI (dBm): -33.00
SNR (dB): 12.00
Raw Packet: T=STAT,CID=0,MODE=SARV,TX=DIS,H=OK,A=0,RSSI=-33.00,SNR=12.00
Test Metrics
Total Commands: 0 Total Responses: 1 Mode Changes: 0 TX Changes: 0 Matches: 0 Mismatches: 0
Exit
Event Log
Ground HMI started.
GROUND RX | Type=Status | CID=0 | Mode=SARVideo | TX=Disabled | Health=OK | Alarm=0 | RSSI=-33.00 dBm | SNR=12.00 dB | Latency=s | Match=NO

# Air Radar HMI



# Drive Test

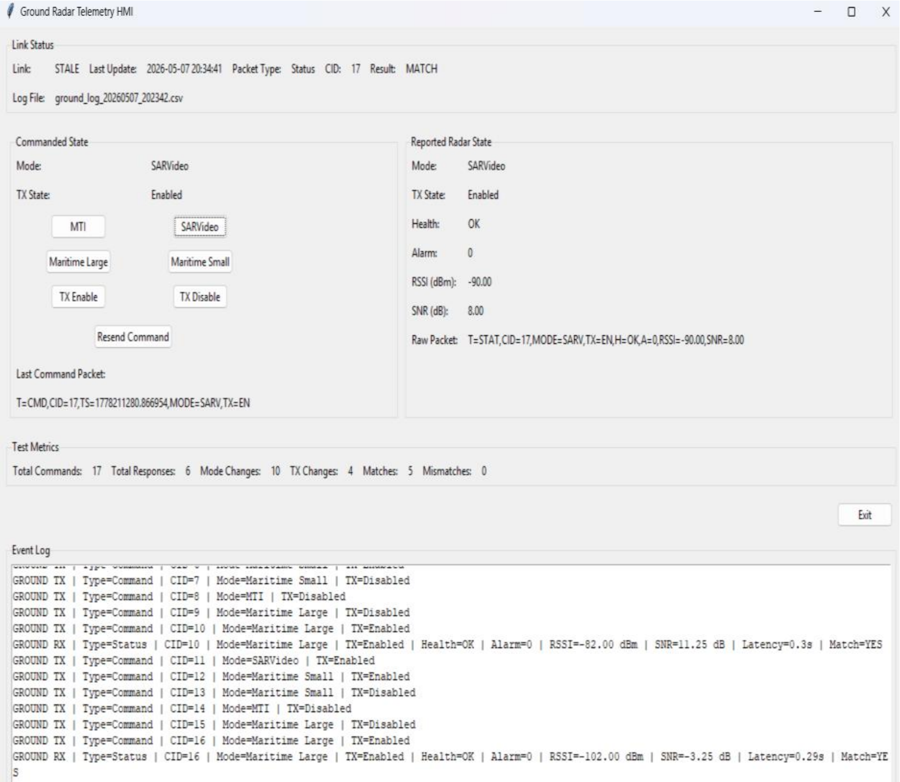
**Method**

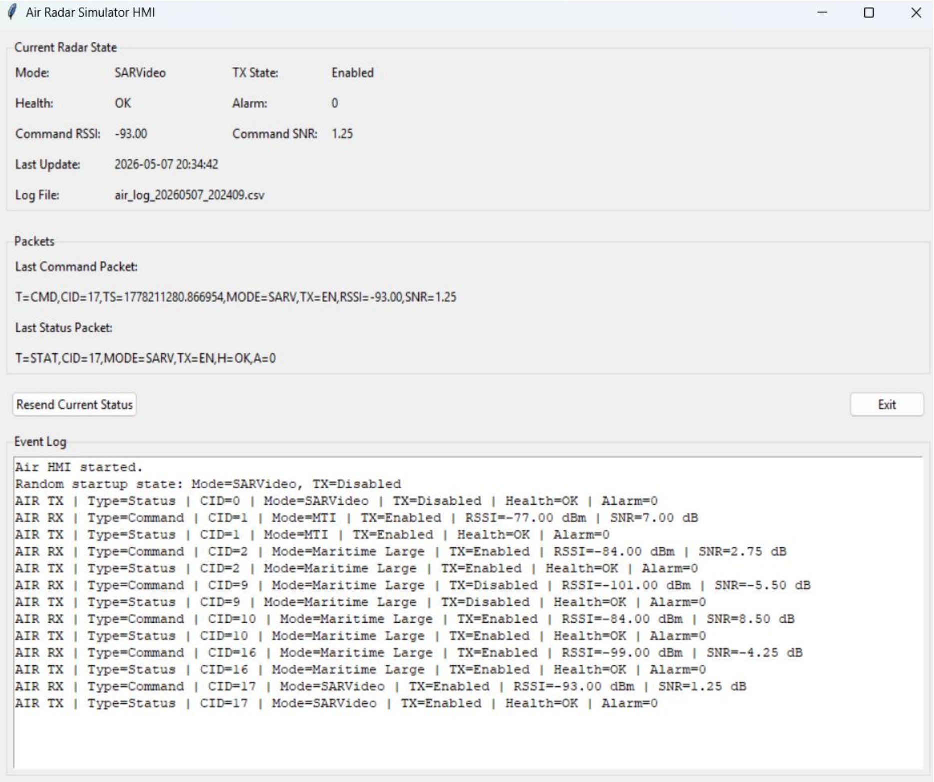
- Speed: 10–15 mph
- Route: around the block
- Trials: 2
- Check: command and returned status match

**Results**

- 100% match to 60 yards
- Failed beyond 60 yards and lost packets
- Same result in both trials
    - 17 Mode changes
    - 6 total response
    - 11 total lost packets

# Drive Test Results from Ground and Air HMI



# MVP Status and Next Steps

**MVP Status**

- The MVP is working
- The ground laptop can send commands to the air laptop
- The air laptop can return the updated radar status
- Both HMIs display and log the test data
- Bench testing and driving tests have been completed

**Next Steps – May9th to June 5th**

- Repeat the driving test and collect more data
- Compare packet success, latency, RSSI, and SNR
- Identify why the link drops after about 60 yards during motion
- Visualize trial data by generating graphs and tables from saved CSV files.
- Finish the final report, GitHub updates, and video presentation